Connecting coverage criteria to explainability

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Why Connect Coverage and Explainable RL?

CSAC coverage criteria
Measure which relevant parts of the agent's behavior have been exercised during testing.

- Entropy–reward coverage
- Reward–constraint coverage
- Safety-state coverage
- Safety-action coverage
- Safety-policy coverage

Explainable Reinforcement Learning (XRL)
Studies how to make an RL agent's behavior and decisions understandable.

- What behavior was exercised?
- Why might the agent choose an action?
- Which reward–cost trade-off is involved?
- Which state features influence the decision?

Central connection

Coverage identifies *which behaviors and risk regions were tested*; XRL methods can help explain *why the agent behaved that way*. Coverage alone does not automatically provide a causal explanation.

Qing et al., *A Survey on Explainable Reinforcement Learning*, arXiv:2211.06665.

Coverage Criteria: What Do They Reveal?

Coverage criterion	**Directly reveals**	**Explainability connection**
Entropy–reward C_{ER}	Which entropy and reward-value regions are visited	Where does the policy sacrifice some reward for exploration?
Reward–constraint C_{RC}	Which reward–cost value combinations are exercised	Where does the policy incur some cost for greater rewards?
Safety-state (CSS)	Which safety-margin regions are visited	Organizes explanations by safe, near-boundary, boundary, and unsafe states
Safety-action (CSA)	Which safe, critical, and unsafe action regions are exercised	When/why does the policy select unsafe actions?
Safety-policy	The policy's probability mass on critical or unsafe actions	Quantifies how much probability the policy assigns to risky actions in a state

Mapping CSAC Coverage to the XRL Survey

The XRL survey groups explanation approaches into four broad categories. The following mapping shows how they can complement the proposed coverage criteria.

XRL category	Relevant CSAC coverage	Potential explanation
Agent-model explanation	Safety-action and safety-policy coverage	Explain why a particular action was selected, or why risky actions can sometimes be selected
Reward explanation	Reward-constraint / Entropy-reward coverage	Explain how Q_C, Q_R, H compete and their tradeoffs.
State explanation	Safety-state coverage	Identify state features associated with safety margins or unsafe transitions.
Task explanation	Trajectory-level analysis across coverage regions	Explain how decisions contribute to a sequence of events or a task stage. This requires an additional task/trajectory analysis layer.

Key point: the coverage criteria locate important cases for explanation; additional XRL methods are needed to generate and validate explanations.

Qing et al., XRL survey.

Example: High Reward, High Constraint Cost

What coverage tells us
A visited state–action region has:

$$Q_R(s, a) \text{ high}, \qquad Q_C(s, a) \text{ high}.$$

This is a critical trade-off region under reward–constraint coverage.
Safety-policy coverage can additionally report:

$$P_{\text{unsafe}}(s)$$

and probability mass assigned to critical actions.

What XRL can add

- **Counterfactual:** what alternative action might reduce cost, and what reward may be lost?
- **Feature attribution:** which observed state features influence the policy or value estimate?
- **Reward/cost analysis:** how do Q_R, Q_C, and λ relate to the selected action?

Post-hoc Explainability for CSAC

Post-hoc explanation analyzes a trained agent's behavior without necessarily changing its policy architecture.

1. **Log each decision/transition:**

$$s_t,\ a_t,\ r_t,\ c_t,\ Q_R(s_t, a_t),\ Q_C(s_t, a_t),\ \mathcal{H}(\pi),\ \alpha,\ \lambda$$

2. **Use coverage to select cases:** prioritize boundary states, unsafe-action selections, and high-reward/high-cost regions.
3. **Generate explanations:** counterfactual alternatives, state-feature attribution, or reward–cost decomposition.
4. **Validate explanations:** check whether they reflect model behavior, remain stable, and help interpret safety-relevant decisions.